

Imaging of Actinomycosis in Various Organs: A Comprehensive Review¹

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ONLINE-ONLY SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe the microbiology, pathogenesis, pathology, and clinical manifestations of actinomycosis.
- Recognize various imaging features of actinomycosis according to the involvement sites.
- Discuss treatment strategy and prognosis for patients with actinomycosis.

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TEACHING POINTS

See last page

Actinomycosis is a chronic suppurative bacterial infection caused by *Actinomyces* species. *Actinomyces israelii* is the organism most commonly found in human disease. Actinomycosis usually manifests with abscess formation, dense fibrosis, and draining sinuses. The disease is further characterized by the tendency to extensively spread beyond normal fascial and connective tissue planes. Actinomycosis occurs most commonly in the cervicofacial region (50%–65%), followed by the thoracic (15%–30%) and abdominopelvic (20%) regions, but rarely involves the central nervous system. Most cases of cervicofacial actinomycosis are odontogenic in origin. In the acute form, cervicofacial disease can manifest with soft-tissue swelling, a painful pyogenic abscess, or a mass lesion. In the subacute to chronic form, a painless indurated mass can spread to the skin, leading to draining sinus tracts. Thoracic manifestations include parenchymal, bronchiectatic, and endobronchial actinomycosis. At computed tomography, pulmonary actinomycosis usually appears as chronic segmental airspace consolidation containing necrotic low-attenuation areas with peripheral enhancement. Abdominopelvic actinomycosis preferentially involves the ileocecal region, ovary, and fallopian tube. The imaging findings favoring abdominopelvic actinomycosis include strong enhancement in the solid portion of the mass after contrast material administration, small rim-enhancing abscesses within the mass, and extensive inflammatory extensions. Actinomycosis in the central nervous system may produce brain abscess, meningitis, subdural empyema, actinomycetoma, and spinal and cranial epidural abscess. In general, actinomycosis responds well to antibiotic therapy, but long-term follow-up after treatment is needed because of frequent relapses.

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Introduction

Actinomycosis is a chronic suppurative disease caused by *Actinomyces* species, which are members of the endogenous flora normally encountered in the mouth, gastrointestinal tract, and female genital tract (1). Actinomycosis usually manifests with several features, such as abscess formation, dense fibrosis, and draining sinuses (2). The clinical and radiologic findings of actinomycosis are variable, depending on the primary site of involvement and the duration of the disease. Indeed, it can frequently resemble other inflammatory and neoplastic lesions, so actinomycosis raises a diagnostic challenge, especially when the clinical symptoms are not typical. Therefore, an understanding of the various imaging findings of actinomycosis grouped by the sites of involvement is necessary for early diagnosis and initial proper antibiotic therapy, to prevent extensive local tissue destruction and minimize unnecessary surgery.

Among several pathogenic *Actinomyces* species in humans, *Actinomyces israelii* is the most common causative agent of actinomycosis (1). The organism is usually unable to cross the mucosal barrier. However, individuals with poor orodental hygiene, those with prolonged use of intrauterine devices, and those receiving bisphosphonate therapy are at high risk of actinomycosis (1).

Actinomycotic infection should be a diagnostic consideration in the following three clinical circumstances: (a) a series of manifestations including chronicity, extensive propagation across tissue planes, and firm to hard masslike features, which are frequently confused with malignant disease; (b) drainage of an abscess by a sinus tract, which may spontaneously close and reform elsewhere; and (c) temporary improvement after a short course of antibiotic treatment, followed by frequent relapses (1). The multiorgan involvement of actinomycosis can be effectively classified as cervicofacial, thoracic, abdominopelvic, and central nervous system disease (1). The frequency of occurrence of actinomycosis according to the sites of potential involvement is shown in the Table (1–8).

The purpose of this article is to familiarize radiologists with the clinical and radiologic characteristics of actinomycosis. First, the microbiology of actinomycosis and its pathogenesis and histopathology are reviewed. Then the clinical manifestations are described, with illustrations of a wide spectrum of radiologic findings in patients with actinomycosis involving the head and neck, the thorax, the abdomen and pelvis, and the central nervous system. Finally, the treatment strategy and the prognosis of actinomycosis are described.

Frequency of Occurrence of Actinomycosis by Site of Potential Involvement

Site	Prevalence (%)
Cervicofacial	50–65
Thoracic	15–30
Abdominopelvic	20
Central nervous system	2–3
Musculoskeletal	Rare
Disseminated	Rare

Sources.—References 1–8.

Microbiology

The causative agents of actinomycosis are gram-positive non-spore-forming anaerobic or microaerophilic bacterial species (2) with a tendency to produce branching filaments that usually fragment into longer or shorter rods, with or without branching (9). *Actinomyces* species were once thought to be fungi because of their branching filaments. However, these species are pleomorphic and branching filamentous bacteria that are currently classified in the genus *Actinomyces* of the order Actinomycetales (2). The genus *Actinomyces* has been expanded considerably in recent years (10). Although several *Actinomyces* species could be pathogenic in humans—including *Actinomyces naeslundii*, *Actinomyces odontolyticus*, *Actinomyces viscosus*, *Actinomyces meyeri*, and *Actinomyces gerencseriae* (1)—actinomycosis is most commonly caused by *A israelii* (1). Although the definite diagnosis is made by microbiologic identification of one of the organisms of actinomycosis from a sterile site, the isolation of *Actinomyces* species is usually achieved in fewer than 50% of cases (3,11). Because most actinomycotic infections are polymicrobial in nature (9,12), a variety of other microorganisms are frequently found at infection sites in addition to *Actinomyces* species.

Pathogenesis and Histopathology

In the pathogenesis of actinomycosis, disruption of the mucosal barrier plays a central role (1). Thus, this disease usually follows conditions that may allow bacterial entry into deep tissues across the mucosal barrier, such as trauma, surgery, or foreign bodies. Typically, a densely fibrotic lesion is one of the features of actinomycosis, which tends to extensively spread beyond fascial and connective tissue planes (1). As the lesion progresses, it is accompanied by abscess formation centrally, which eventually leads to sinus tracts extending from the abscess to the skin or adjacent organs (1).

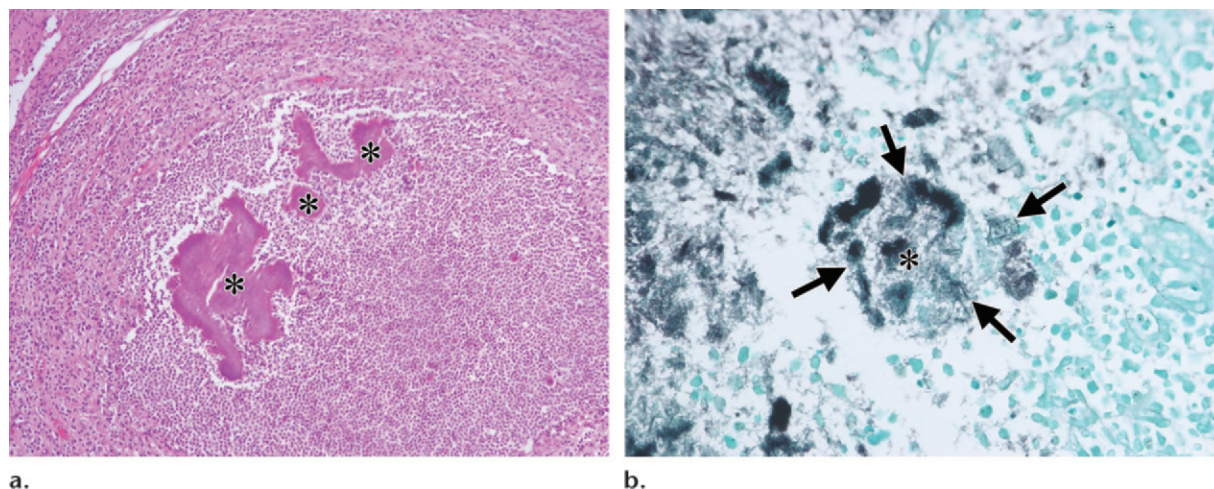


Figure 1. Sulfur granules in the surgical specimen obtained from a 50-year-old woman. **(a)** Photomicrograph (original magnification, $\times 100$; hematoxylin-eosin stain) shows sulfur granules (*) that are round, oval, or horseshoe-shaped basophilic masses with a radiating fringe of eosinophilic clubs. **(b)** Photomicrograph (original magnification, $\times 400$; Gomori methenamine-silver stain) depicts filamentous bacteria (arrows) in a sulfur granule (*).

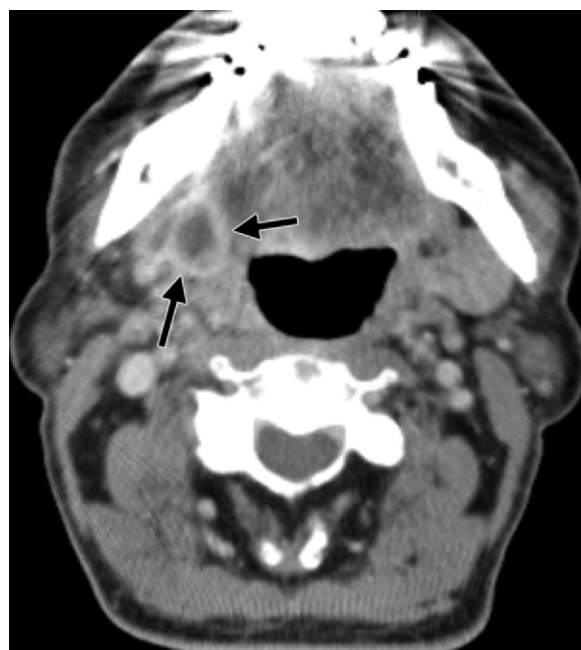


Figure 2. Acute cervicofacial actinomycosis in a 63-year-old man. Axial contrast material-enhanced computed tomographic (CT) image shows a peripheral ring-enhancing lesion (arrows) with central suppurative necrosis in the right submandibular space.

From a diagnostic perspective, actinomycosis is further characterized by the histopathologic finding of “sulfur granules.” The presence of these sulfur granules in pus from the abscess, in exudates from a sinus tract, or in a tissue specimen is the single most helpful histopathologic finding in making a definite diagnosis of actinomycosis. The sulfur granules, which are named for their yellow color, represent tangled filaments of *Actinomyces* species (2). Those granules are

comprised of bacterial elements and tissue debris (4). With hematoxylin-eosin staining, sulfur granules are seen as round, oval, or horseshoe-shaped basophilic masses with a radiating fringe of eosinophilic clubs (Fig 1a). Given that sulfur granules may occur in infectious diseases other than actinomycosis, such as nocardiosis and eumycetoma (2), special stains such as Gomori methenamine-silver stain may be needed to show that the sulfur granules are composed of branching bacteria and not fungi, cocci, or bacilli (Fig 1b) (1).

Clinical Manifestations

Cervicofacial Disease

Actinomycosis most commonly involves the cervicofacial region, which represents more than 50% of cases (3). Most cervicofacial *Actinomyces* infections are odontogenic in origin and develop mainly in individuals with intact immune systems (5). The risk factors for cervicofacial infections include poor dental hygiene, recent dental extraction, dental caries, oromaxillofacial trauma, chronic tonsillitis, otitis, and mastoiditis (2,6). In addition, osteonecrosis caused by radiation therapy or by bisphosphonate therapy has been recently associated with actinomycotic infection of the mandible and maxilla (13). The most commonly involved site is the angle of the jaw or the submandibular region, followed by the cheek, the submental space, the masticator space, and the temporomandibular joint (1).

In the acute form, cervicofacial disease can manifest with soft-tissue swelling, a painful pyogenic abscess, or a mass lesion (Fig 2). In the subacute to chronic form, a painless indurated mass can spread to the skin and is often accompanied

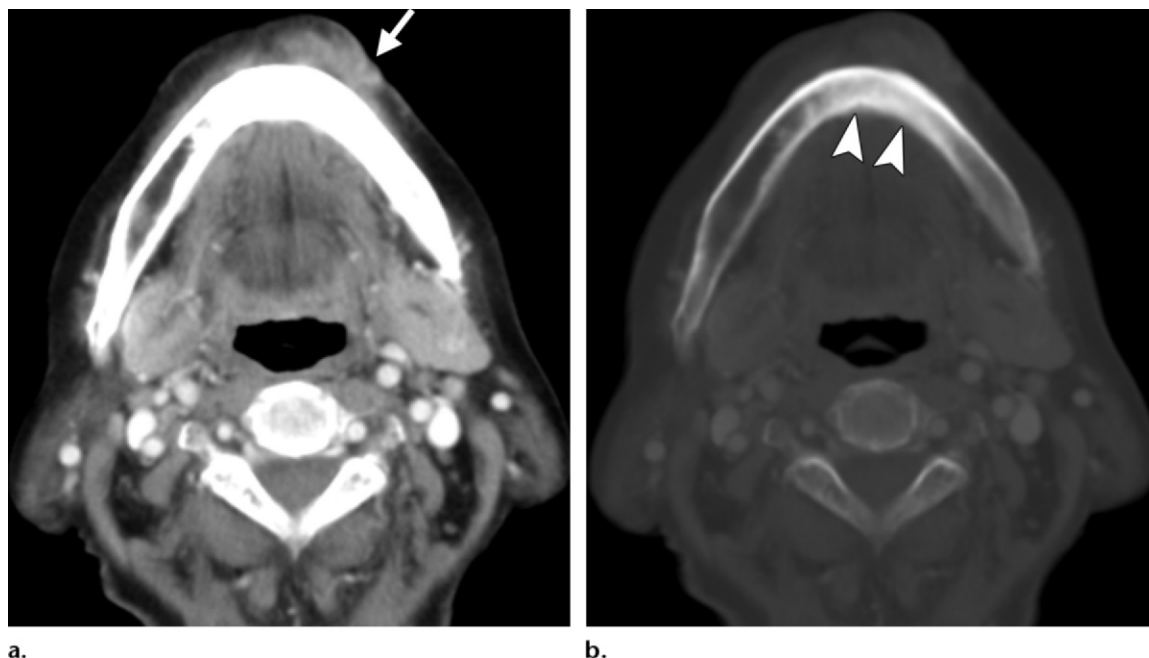


Figure 3. Chronic cervicofacial actinomycosis with a draining sinus tract in a 73-year-old woman. **(a)** Axial contrast-enhanced CT image shows an elliptical mass with a focal skin defect (arrow), which corresponds to the cutaneous fistula. **(b)** Axial contrast-enhanced CT image (bone window) depicts focal sclerotic change (arrowheads) in the adjacent mandible, a finding suggestive of osteomyelitis.

by draining sinus tracts (Fig 3) (2). Although actinomycotic infection can be associated with osteomyelitis in the adjacent mandible or maxilla, this association is not commonly observed (14). In some cases, an actinomycotic lesion is seen as a mass with relatively homogeneous enhancement that has small necrotic areas or cystic portions (Fig 4). This pattern of enhancement might be attributed to the abundant granulation tissue and intense fibrosis (15). Maxillary involvement occurs less frequently; with this type of involvement, a soft-tissue lesion associated with osteomyelitis and cutaneous or maxillary sinus fistulas can be observed (Fig 5).

Actinomycosis tends to spread into the adjacent neck space, ignoring normal fascial planes (Fig 6) (1,6,15). **This infiltrative nature may be explained by the action of the proteolytic enzymes of the organism (16).** Because *Actinomyces* species usually do not spread lymphatically as a result of the size of the organism, actinomycosis is rarely associated with regional lymphadenopathy (6,15,17). Thus, in distinguishing actinomycosis from a malignancy, the diagnosis of actinomycosis should be considered if a highly infiltrative contrast material-enhanced mass is seen that is accompanied by no or a few reactive regional lymph nodes.

Thoracic Disease

Actinomyces species may cause bronchopulmonary infection, as well as pleural, chest wall, or mediastinal involvement; and the thoracic form



Figure 4. Cervicofacial actinomycosis in a 49-year-old woman. Coronal contrast-enhanced CT image shows a strongly enhanced mass (arrows) in the left submandibular gland. Several small nonenhanced areas are noted within the mass, findings suggestive of small abscesses. Soft-tissue infiltration to the adjacent fat plane and fascial thickening (arrowheads) are also depicted.

of the disease accounts for 15%–30% of the cases of actinomycosis (12,18). Thoracic actinomycosis is believed to spread into the thoracic organs mainly through aspiration of organisms

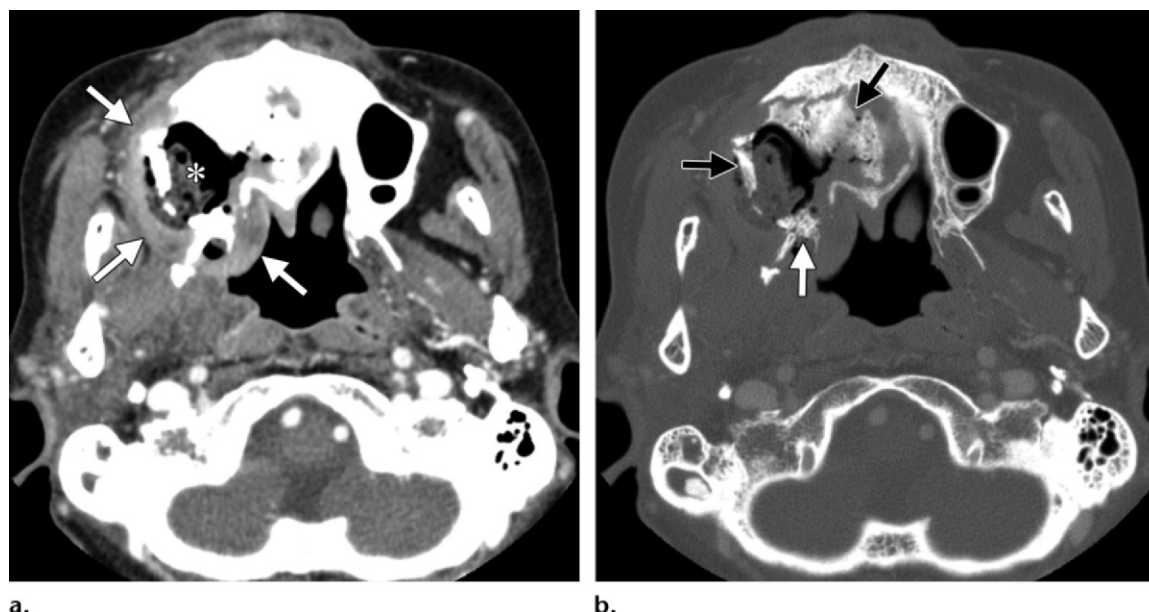


Figure 5. Actinomycosis involving the maxilla in a 75-year-old woman. **(a)** Axial contrast-enhanced CT image shows an ill-defined and poorly enhanced lesion (*) with air bubbles and adjacent soft-tissue thickening (arrows) in the right maxilla and maxillary sinus. **(b)** Axial contrast-enhanced CT image (bone window) shows irregular bone destruction and diffuse sclerotic change (arrows), findings that indicate chronic osteomyelitis.

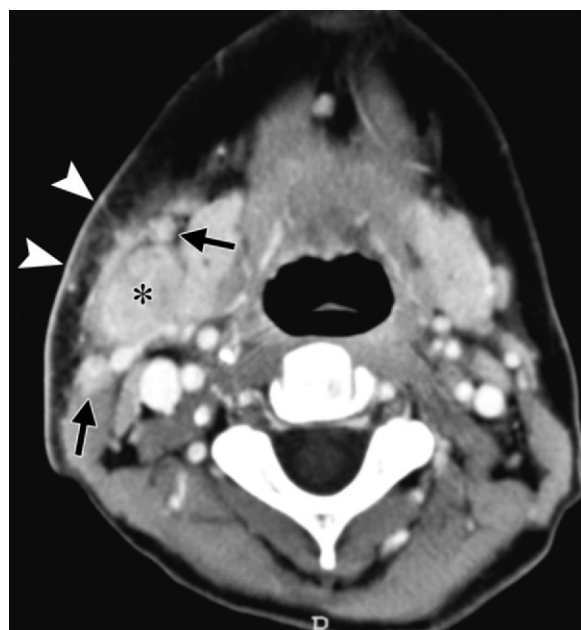


Figure 6. Cervicofacial actinomycosis in a 53-year-old woman. Axial contrast-enhanced CT image shows a homogeneous and strongly enhanced mass (*) without a necrotic or cystic portion in the right submandibular space. Regional reactive lymph nodes (arrows) and inflammatory spread to the skin (arrowheads) are also depicted.

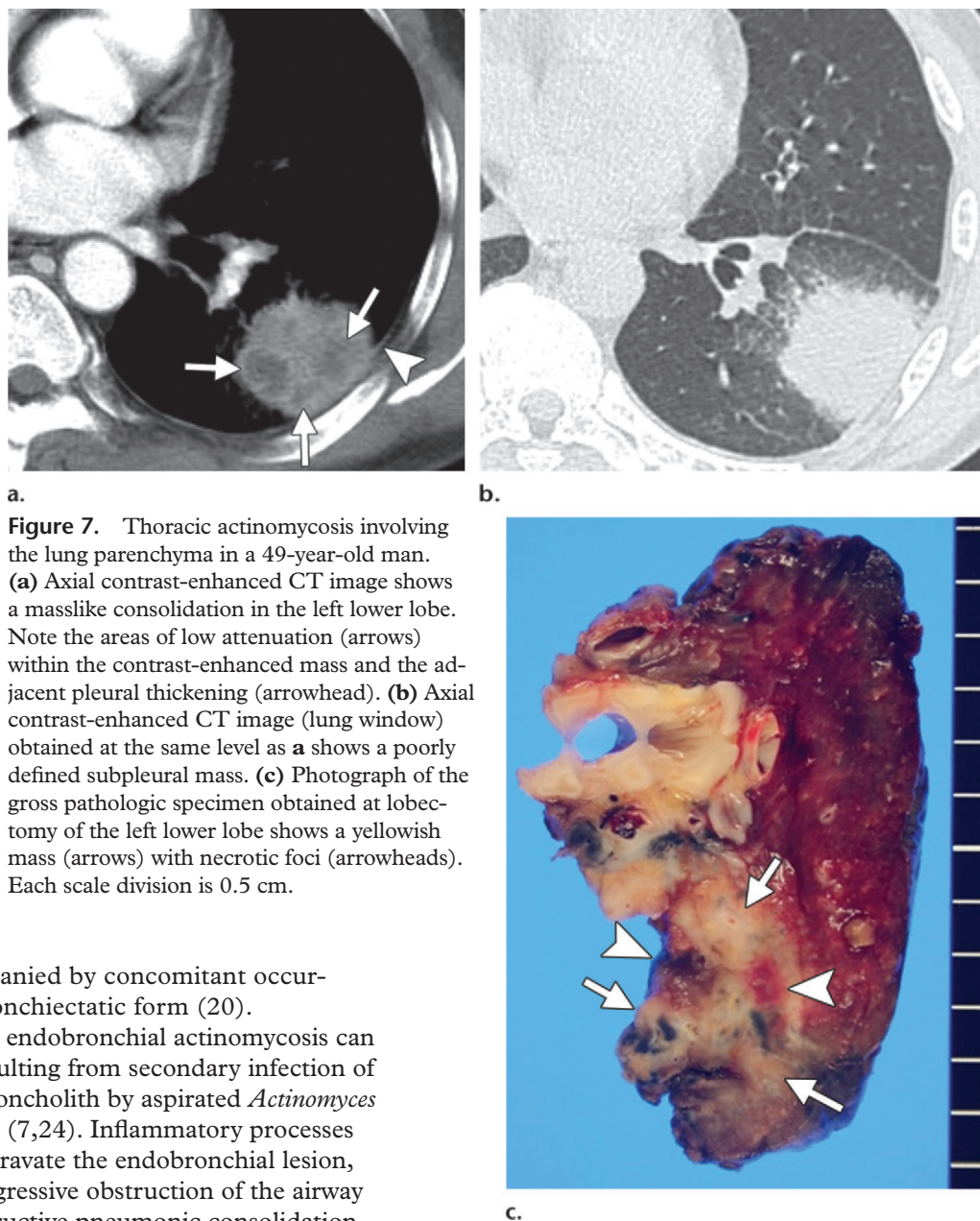
from the oropharynx (19). However, the thoracic form of the disease could also result from (a) direct extension into the mediastinum from cervicofacial infection, (b) transdiaphragmatic or retroperitoneal spread from the abdomen,

or, rarely, (c) hematogenous spread to the lung (2). Thoracic infection may occur more frequently in patients with underlying pulmonary disease, such as emphysema, chronic bronchitis, or bronchiectasis, and those with previous lung damage (20).

Pulmonary actinomycotic infection usually manifests with a mass or pneumonia (1), which predominantly affects the lower lobes (21). **At CT, pulmonary actinomycotic infection usually appears as chronic segmental airspace consolidation containing necrotic low-attenuation areas with peripheral enhancement, often accompanied by adjacent pleural thickening (Fig 7) (19,22).** Parenchymal actinomycosis is associated with pleural involvement, that is, pleural thickening, effusion, or empyema, in more than 50% of cases (1). Multiple small cavities may develop within the parenchymal lesion (Fig 8). In addition to parenchymal actinomycosis, other forms of pulmonary actinomycosis have been reported.

Given that *Actinomyces* species tend to invade devitalized tissue (23), it is not surprising that patients with parenchymal destruction and bronchiectasis caused by prior tuberculosis or other infections are predisposed to secondary actinomycotic infection. In general, CT findings of the bronchiectatic form of actinomycosis include localized areas of bronchiectasis, irregular bronchial wall thickening, and irregular peribronchial consolidation with or without necrotic foci (Fig 9) (22). Parenchymal actinomycosis

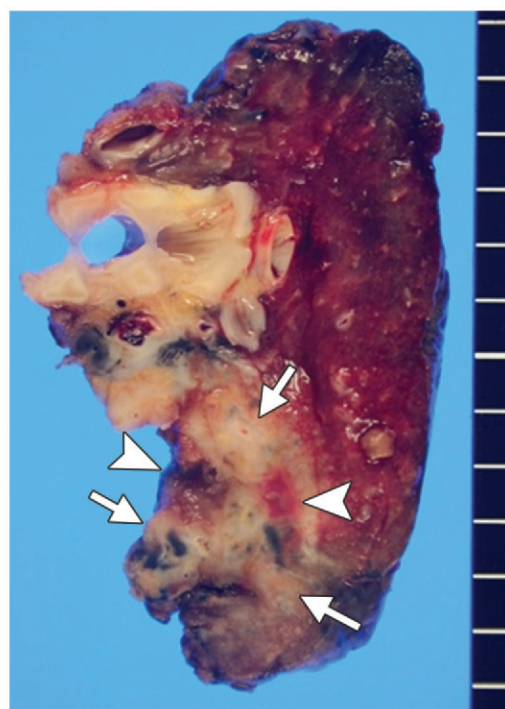
Teaching
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a.

Figure 7. Thoracic actinomyces involving the lung parenchyma in a 49-year-old man. (a) Axial contrast-enhanced CT image shows a masslike consolidation in the left lower lobe. Note the areas of low attenuation (arrows) within the contrast-enhanced mass and the adjacent pleural thickening (arrowhead). (b) Axial contrast-enhanced CT image (lung window) obtained at the same level as a shows a poorly defined subpleural mass. (c) Photograph of the gross pathologic specimen obtained at lobectomy of the left lower lobe shows a yellowish mass (arrows) with necrotic foci (arrowheads). Each scale division is 0.5 cm.

b.



c.

may be accompanied by concomitant occurrence of the bronchiectatic form (20).

In rare cases, endobronchial actinomyces can occur, likely resulting from secondary infection of a preexisting broncholith by aspirated *Actinomyces* species (Fig 10) (7,24). Inflammatory processes may further aggravate the endobronchial lesion, resulting in progressive obstruction of the airway with distal obstructive pneumonic consolidation. In most cases of the endobronchial form of actinomyces, a calcified endobronchial nodule, that is, a broncholith, was reported to be caused by a previous tuberculous infection (7). Thus, when a central endobronchial calcified nodule (broncholith) associated with distal postobstructive pneumonia is depicted at CT in a region with endemic tuberculosis, the differential diagnosis needs to include endobronchial actinomyces.

Pulmonary actinomyces may extend to the mediastinum and chest wall across anatomic barriers, such as fissures and pleura. However, mediastinal actinomyces is known to be extremely rare (1).

Abdominopelvic Disease

On average, about 20% of the patients with actinomyces present with the abdominopelvic form

of the disease (25). Abdominal actinomyces usually occurs after the disruption of the intestinal mucosal integrity caused by appendicitis, colonic diverticulitis, penetrating trauma, or surgery of the gut (26). However, on occasion, the inciting conditions may not be identified. Abdominal actinomyces can remain undetected for months to years owing to its indolent nature and nonspecific symptoms (2). Actinomycotic infection preferentially involves the ileocecal region of the bowel and extends to surrounding structures across fascial and connective tissue planes (Fig 11). As noted, the aggressive nature of the infiltration is the telltale clue for actinomyces (16,27,28).

In abdominal actinomyces, the bowel is presumed to be secondarily affected, as evidenced by

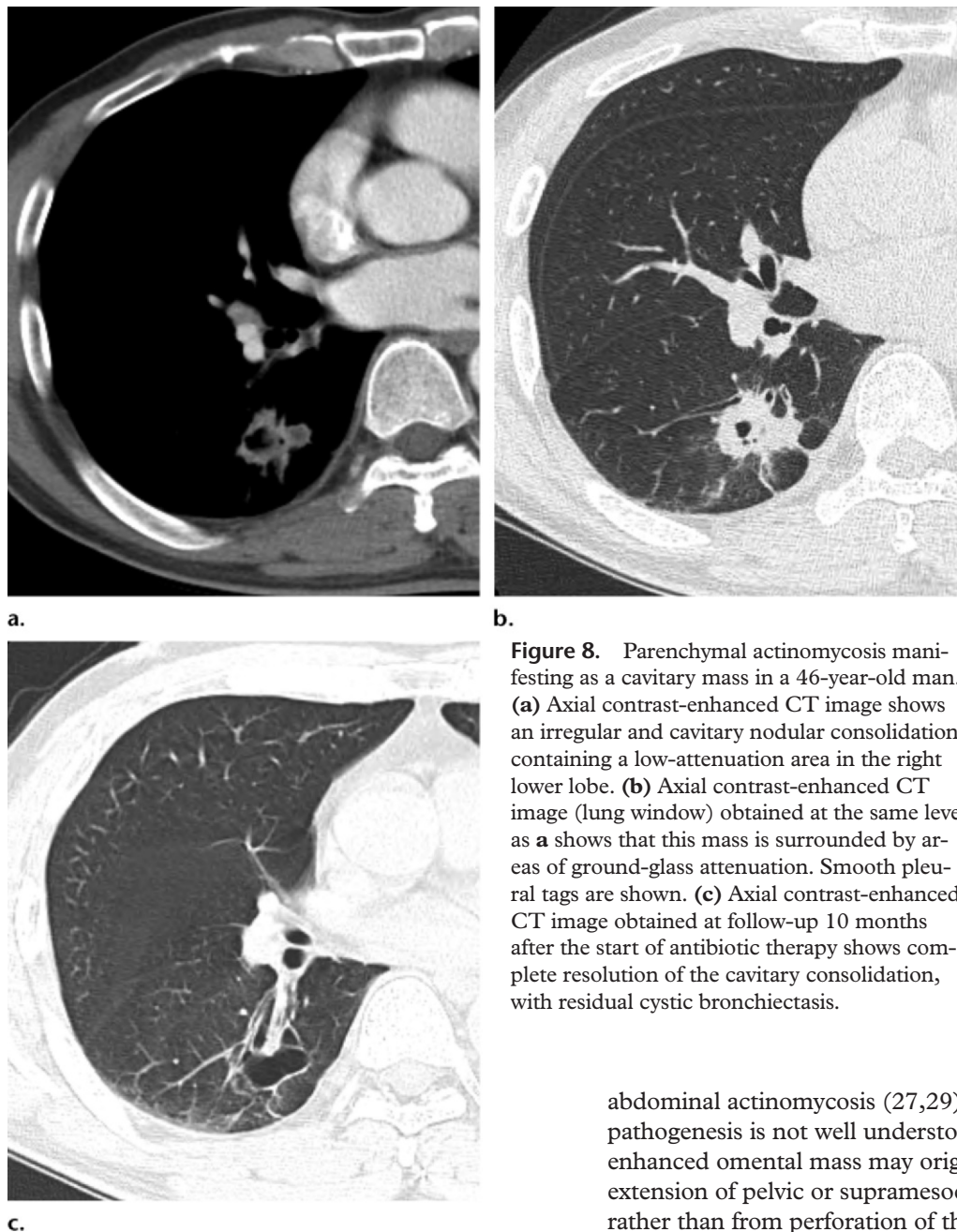


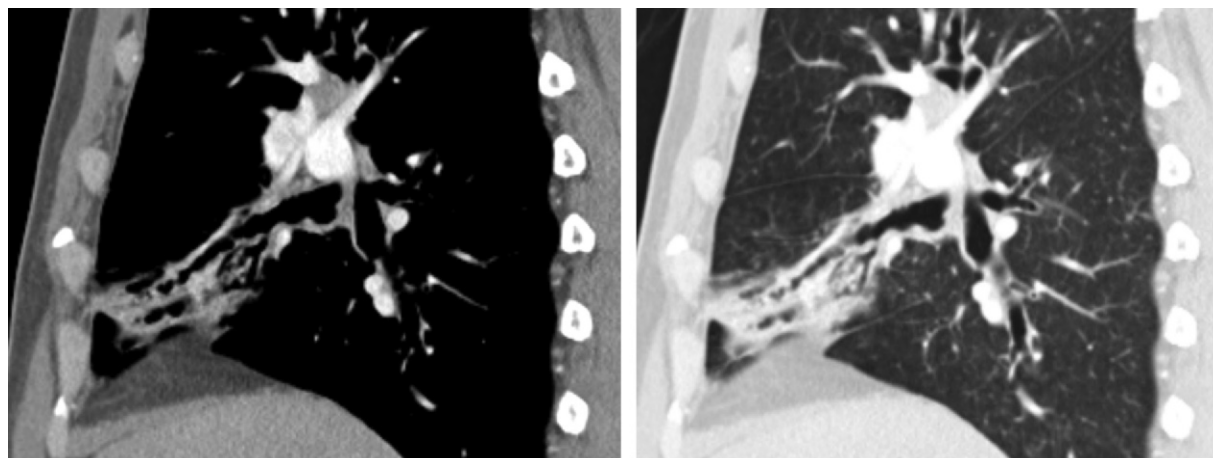
Figure 8. Parenchymal actinomycosis manifesting as a cavitary mass in a 46-year-old man. **(a)** Axial contrast-enhanced CT image shows an irregular and cavitary nodular consolidation containing a low-attenuation area in the right lower lobe. **(b)** Axial contrast-enhanced CT image (lung window) obtained at the same level as **a** shows that this mass is surrounded by areas of ground-glass attenuation. Smooth pleural tags are shown. **(c)** Axial contrast-enhanced CT image obtained at follow-up 10 months after the start of antibiotic therapy shows complete resolution of the cavitary consolidation, with residual cystic bronchiectasis.

a report that mucosal changes were minimal or absent at colonoscopy (27). **In cases of actinomycotic disease involving the bowel, the common CT finding is concentric wall thickening, accompanied by a cystic or solid mass in the vicinity of the affected bowel segment (Fig 12) (27).** This mass is characteristically surrounded by prominent inflammatory infiltrations. Because of the presence of extensive and dense fibrosis, a hallmark of actinomycosis, solid portions of the mass and the surrounding soft-tissue strands around the mass tend to show strong enhancement after contrast material administration. The invasive nature of the mass often is suggestive of a tumor. However, regional lymphadenopathy is rarely associated with

abdominal actinomycosis (27,29). Although the pathogenesis is not well understood, a strongly enhanced omental mass may originate from the extension of pelvic or supramesocolic infection, rather than from perforation of the bowel (Fig 13) (27). Interestingly, despite the unrestricted infiltrative nature of abdominal actinomycosis, it is unlikely to spread into the entire peritoneal cavity. Ascites is usually minimal or absent.

Pelvic actinomycosis is closely associated with the prolonged use of intrauterine devices, usually for more than 2 years (Fig 14) (4,8). The pelvic form of the disease may even occur months or years after the removal of an intrauterine device (30). Pelvic actinomycosis may also result from extension of abdominal actinomycosis, in which the source of infection is usually latent ileocecal disease (2). In pelvic actinomycosis, the ovary and fallopian tube are most commonly affected.

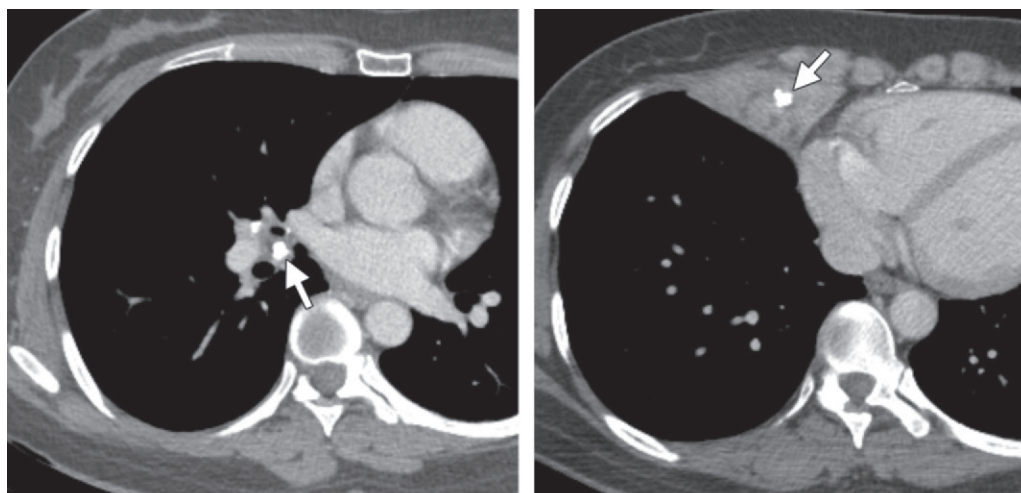
The imaging findings suggestive of tubo-ovarian actinomycosis include strong enhancement in the solid portion of the mass after contrast



a.

b.

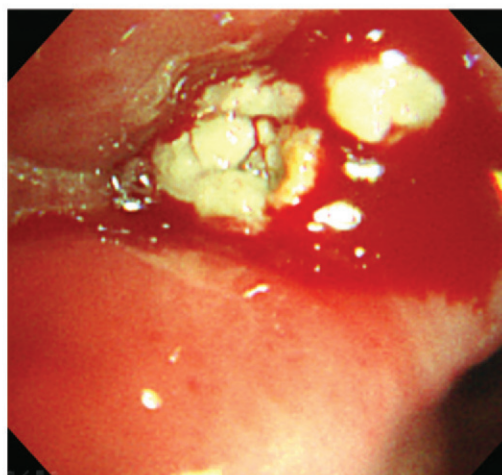
Figure 9. Pulmonary actinomycosis associated with bronchiectasis in a 59-year-old man. Sagittal contrast-enhanced reformatted CT images obtained with mediastinal window settings (**a**) and lung window settings (**b**) show tubular bronchiectasis with peribronchial consolidation and adhesive atelectasis in the right middle lobe.



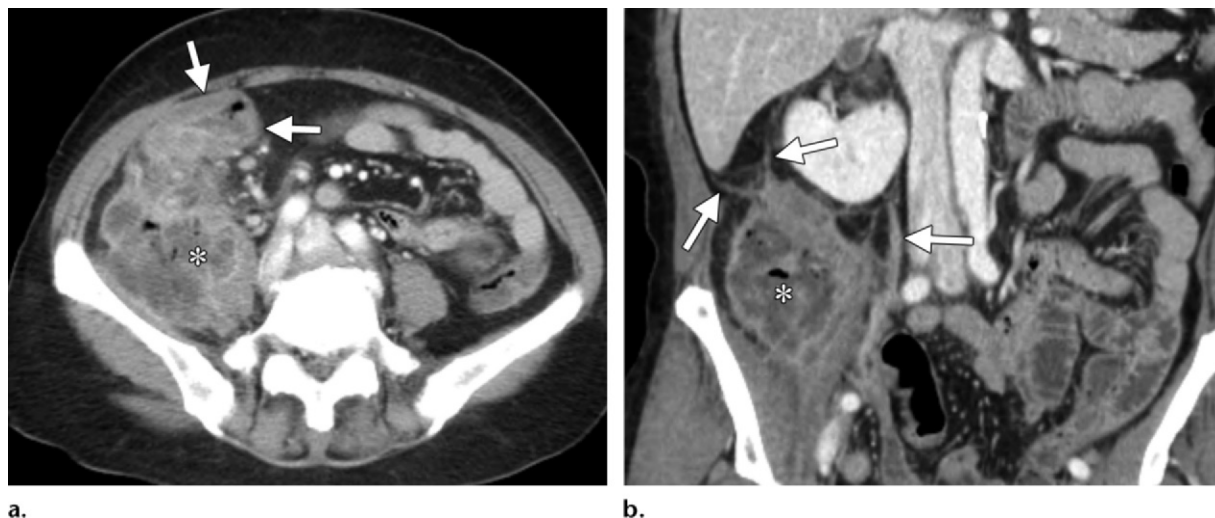
a.

b.

Figure 10. Endobronchial actinomycosis associated with broncholithiasis in a 41-year-old woman. (**a**) Axial contrast-enhanced CT image shows a calcified endobronchial nodule (arrow) obstructing the lumen of the right middle lobar bronchus. Note the surrounding low-attenuation area around the broncholith, a finding suggestive of necrotic material. (**b**) Axial contrast-enhanced CT image obtained at a lower level than **a** shows distal segmental atelectasis containing a fluid-filled bronchus and a calcified nodule (arrow) in the right middle lobe. (**c**) Bronchoscopic image shows yellowish broncholiths with surrounding hemorrhage in the right middle lobar bronchus.



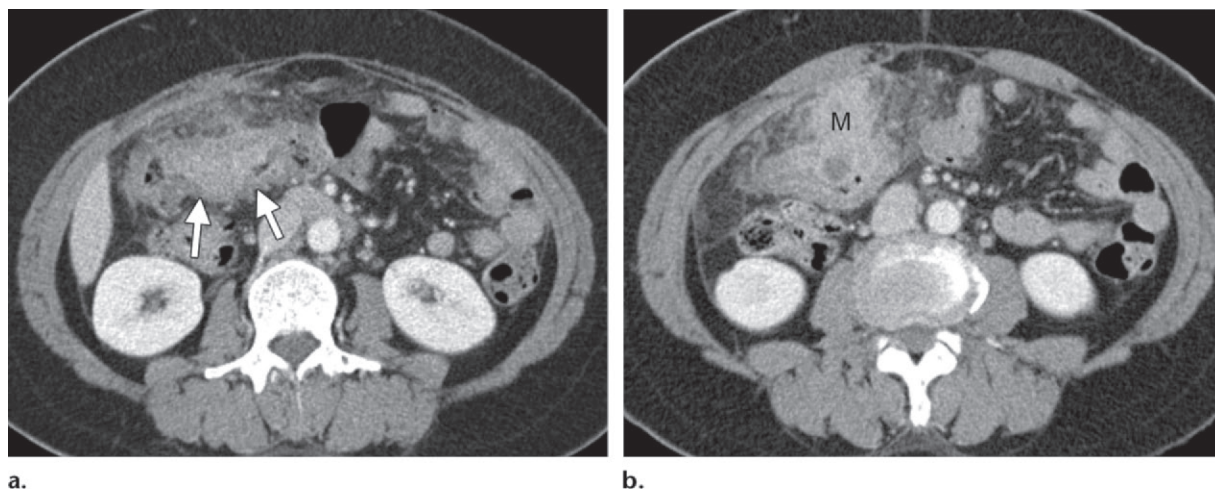
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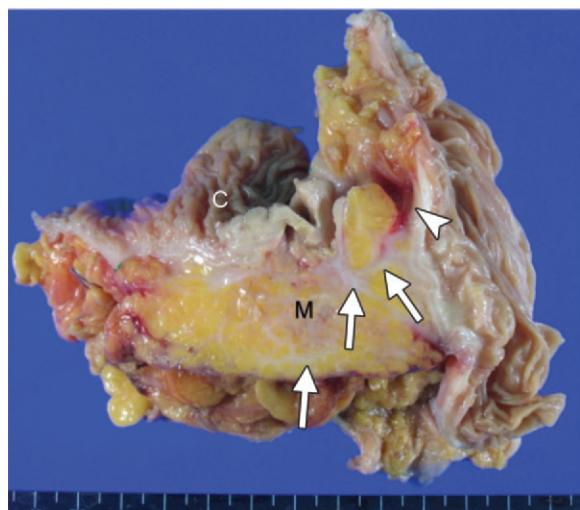
b.

Figure 11. Abdominal actinomycosis involving the cecum in a 64-year-old woman. **(a)** Axial contrast-enhanced CT image shows focal wall thickening of the cecum (arrows) and also a heterogeneously enhancing necrotizing mass (*) with internal air bubbles in the retrocecal area. **(b)** Coronal contrast-enhanced reformatted CT image shows thickening of the retroperitoneal septa and fascia (arrows) around the necrotizing mass (*).



a.

b.



c.

Figure 12. Abdominal actinomycosis involving the transverse colon in a 52-year-old woman. **(a)** Axial contrast-enhanced CT image shows concentric wall thickening and enhancement of the transverse colon (arrows) with adjacent inflammatory infiltration in the greater omentum. **(b)** Axial contrast-enhanced CT image obtained at a lower level than **a** shows a strongly enhanced ill-defined solid mass (*M*) with surrounding fat stranding in the greater omentum. **(c)** Photograph of the gross pathologic specimen shows a yellowish mass (*M*) containing whitish fibrotic bands (arrows). Focal ulceration (arrowhead) is also seen in the transverse colon (*C*). Each scale division is 0.5 cm.

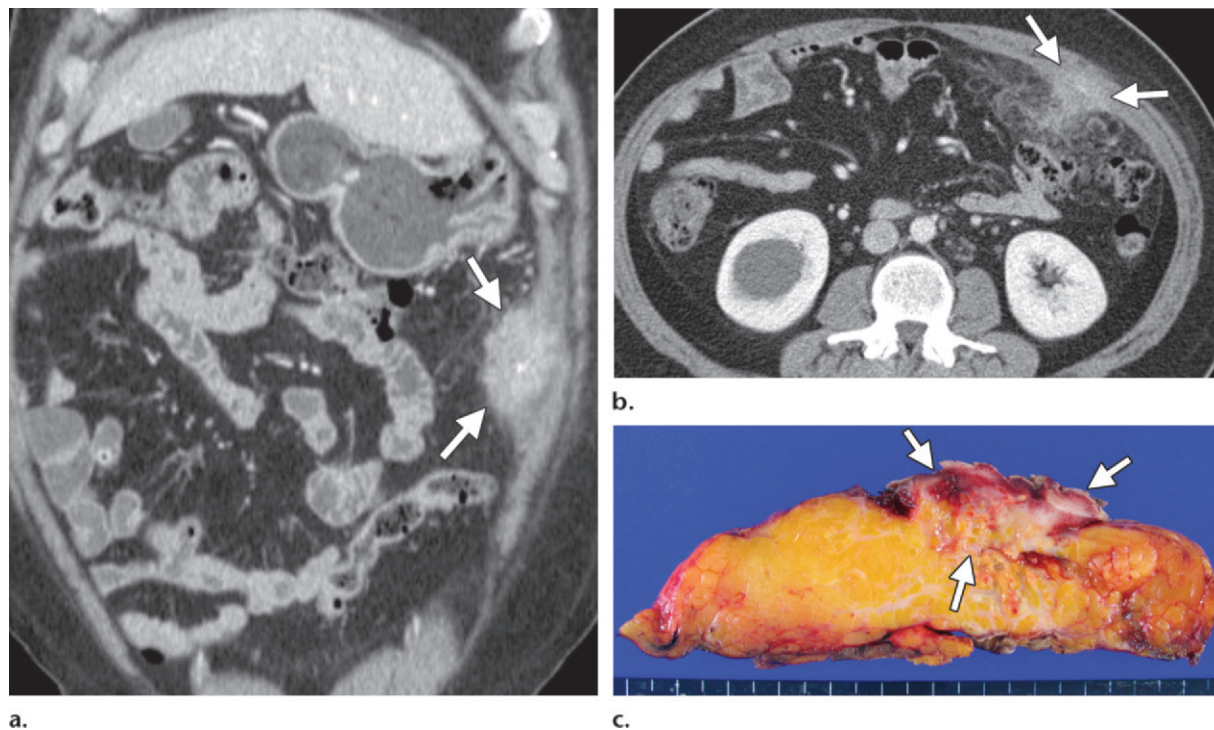


Figure 13. Abdominal actinomycosis with extension to the anterior abdominal wall in a 46-year-old man. **(a)** Coronal contrast-enhanced reformatted CT image shows a strongly enhanced solid mass (arrows) in the greater omentum. **(b)** Axial contrast-enhanced CT image shows that the mass extends to the adjacent anterior abdominal wall (arrows). **(c)** Photograph of the gross pathologic specimen shows an ill-defined mass (arrows) containing rich whitish fibrosis in the omentum. Each scale division is 0.5 cm.

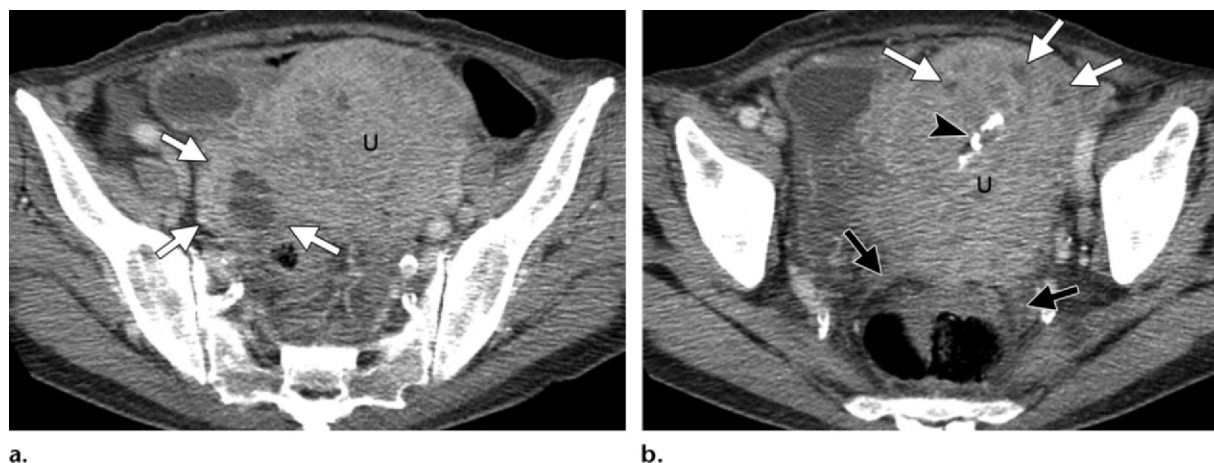
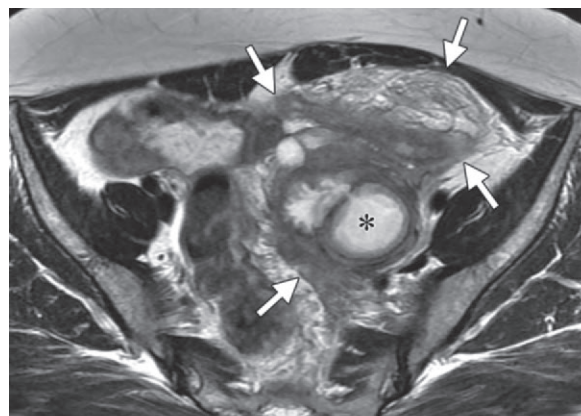


Figure 14. Pelvic actinomycosis involving the adnexa and uterus in a 68-year-old woman. **(a)** Axial contrast-enhanced CT image shows a mixed solid and cystic mass (arrows) in the right adnexa. U = uterus. **(b)** Axial contrast-enhanced CT image obtained at a lower level than **a** shows diffuse enlargement of the uterus (U) with small abscesses (white arrows). Note the perirectal soft-tissue infiltrations (black arrows) and the intrauterine device (arrowhead).

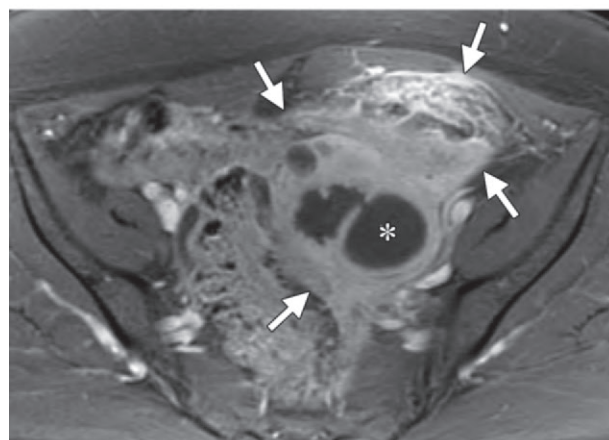
material administration, small rim-enhancing abscesses within the mass, and extensive inflammatory extensions (Fig 15) (31). At CT and MR imaging, tubo-ovarian actinomycosis usually appears to be more solid than the usual tubo-ovarian abscess in the adnexal region (31). Owing to the presence of prominent fibrotic tissue, the inflammatory strandings and the solid component of the

mass tend to demonstrate intermediate to low signal intensity on T2-weighted MR images (31).

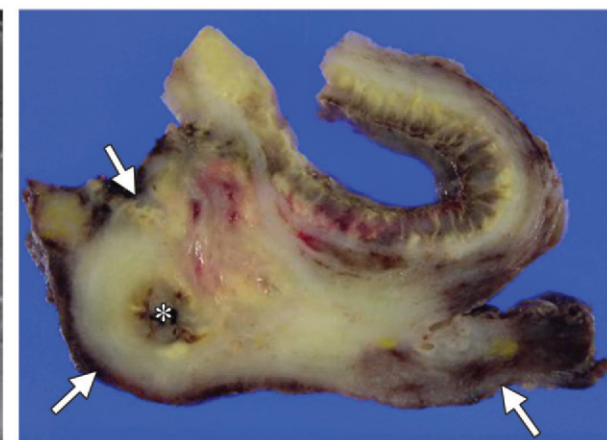
Pelvic actinomycosis can also spread extensively to the uterus, urinary bladder, rectal area, urachus, abdominal wall, and peritoneum (Fig 16) (18,32,33). As pelvic actinomycosis progresses, it can result in a frozen pelvis, resembling pelvic malignancy or endometriosis (1,4).



a.



b.



c.

Figure 15. Tubo-ovarian actinomycosis in a 41-year-old woman. (a) Axial T2-weighted magnetic resonance (MR) image shows a mixed cystic and solid mass (*) in the left adnexa, with marked surrounding inflammatory infiltrations (arrows). (b) Axial contrast-enhanced T1-weighted MR image shows avid enhancement of the surrounding inflammatory infiltrations (arrows), as well as a solid portion in the mass (*). (c) Photograph of the gross pathologic specimen shows a hard yellow to light gray mass (arrows) with necrosis (*) in the left adnexa.



a.



b.

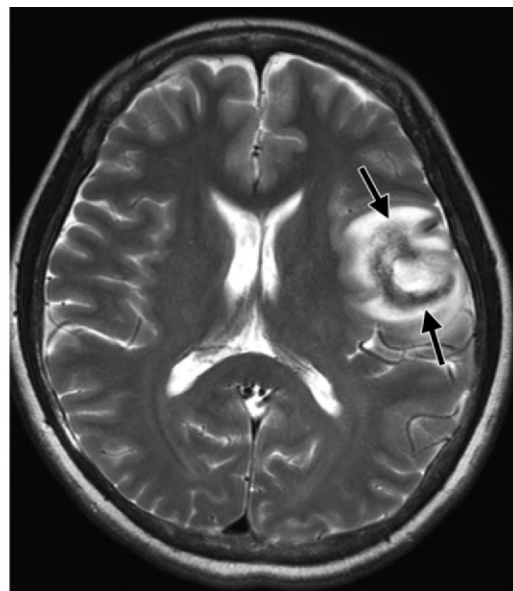
Figure 16. Pelvic actinomycosis involving the urinary bladder and urachus in a 39-year-old woman. (a) Sagittal contrast-enhanced reformatted CT image shows wall thickening of the urinary bladder associated with a strongly enhanced lesion (arrows) extending from the dome of the bladder to the umbilicus. (b) Axial contrast-enhanced CT image shows marked inflammatory fat stranding (arrows) in the anterior pelvic cavity, as well as an intrauterine device (arrowhead).

Central Nervous System Disease

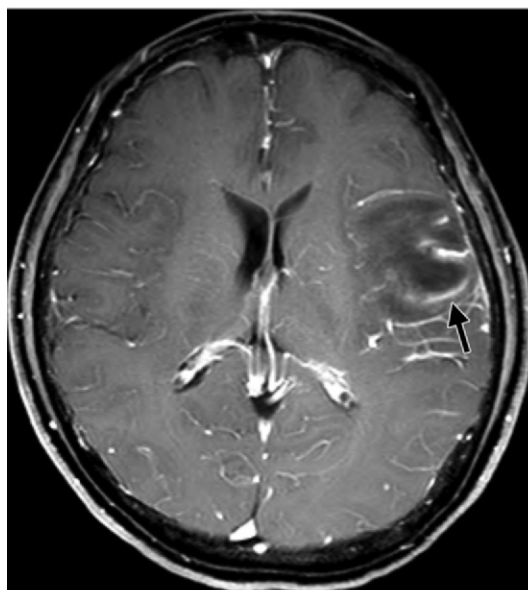
Actinomycosis rarely involves the central nervous system, with infection of this site occurring in about 2%–3% of actinomycotic infections (34,35). Involvement of the central nervous sys-

tem usually results from hematogenous spread from a distant primary site such as the oral cavity, lung, abdomen, or pelvis (36). However, direct extension from foci of infection in the face, jaw, ears, or paranasal sinuses can occur by way

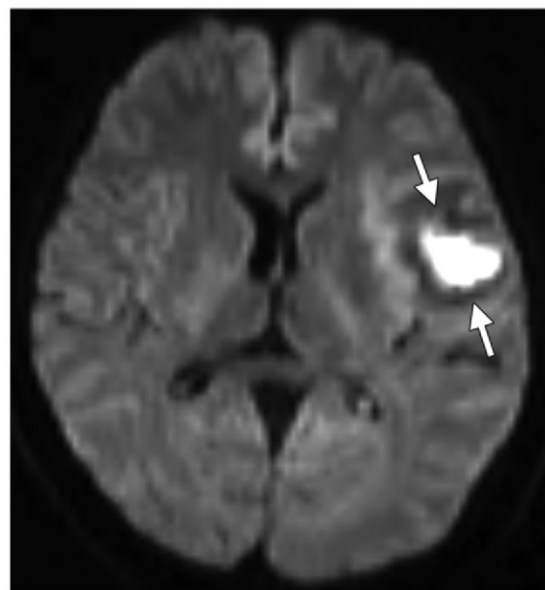
Figure 17. Actinomycotic cerebral abscess in a 39-year-old woman. **(a)** Axial T2-weighted MR image shows a cystic mass (arrows) with a thick low-signal-intensity rim and perilesional edema in the left frontal lobe. **(b)** Axial T1-weighted MR image obtained after administration of gadolinium-based contrast material shows eccentric and thick rim enhancement of the mass (arrow). **(c)** Axial diffusion-weighted MR image shows the central portion of the mass with restricted diffusion and a rim without diffusion restriction at the margin of the mass (arrows), findings that correspond to the nonenhanced central portion and the rim-enhancing peripheral portion of the mass, respectively, in **b**.



a.



b.



c.

of connective tissue planes or foramina at the skull base (35). Although the signs and symptoms mimic those of pyogenic infection, the symptoms are usually mild and indolent; and in turn, the duration of symptoms before diagnosis is typically longer than that for pyogenic brain abscess (2,36,37). Actinomycotic brain infection manifests with fever in fewer than 50% of cases (2). Further, there may be little evidence to imply an infectious disease. This is why an initial diagnosis of a neoplasm is frequently made.

Although brain abscess is reported to be the most common type of central nervous system manifestation of actinomycosis, accounting for about 75% of all central nervous system involvement, actinomycosis may also manifest in the central nervous system as meningitis, subdural

empyema, actinomycetoma, and spinal and cranial epidural abscess (2). Actinomycotic brain abscess is usually single and involves the temporal and frontal lobes (2). The imaging findings include a nonenhanced central portion that represents pus, an enhancing thick wall that may be irregular or nodular, and perilesional edema (Fig 17) (36). These findings also may be noted in other causes of brain abscess and malignancy. However, in contrast with cystic brain tumors, the core of the abscess is depicted as an area with restricted diffusion and the peripheral wall is depicted as an area without diffusion restriction at diffusion-weighted imaging (37,38). Meningeal involvement with actinomycosis may result in chronic meningitis, which is easily confused with tuberculous meningitis (2) because

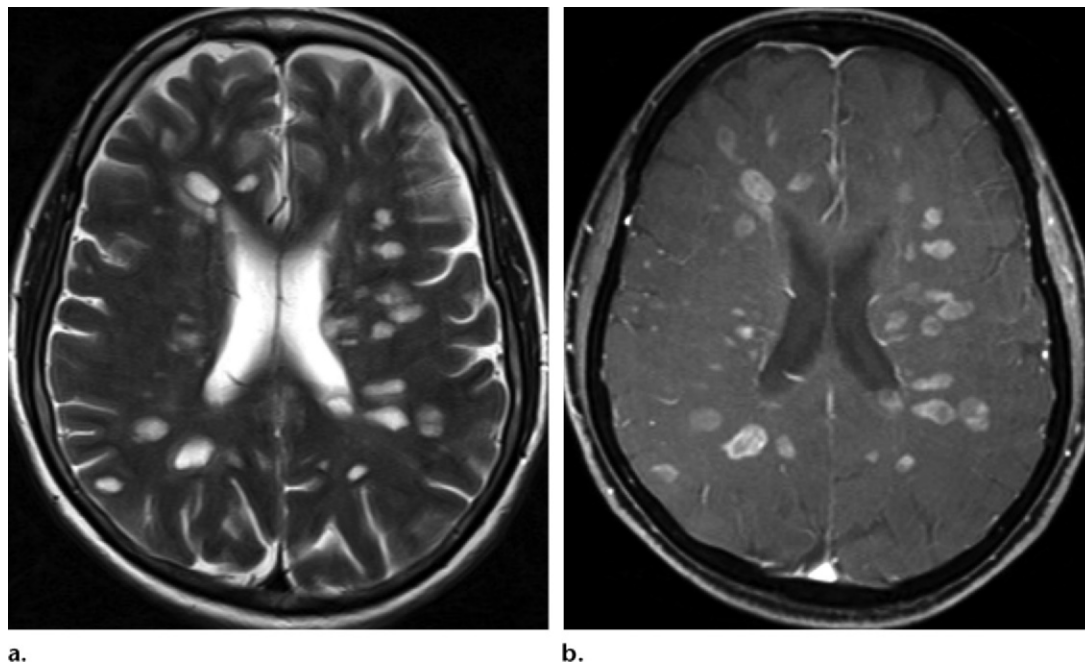


Figure 18. Cerebral actinomycetoma in a 69-year-old woman. **(a)** Axial T2-weighted MR image shows multiple round or oval hyperintense nodular lesions in the periventricular and subcortical white matter. **(b)** Axial gadolinium-enhanced T1-weighted MR image depicts nodular or rim enhancement of the lesions.

of the similar findings in the cerebrospinal fluid (normal or low glucose level, elevated protein level, mononuclear pleocytosis, and negative culture).

Less commonly, actinomycotic infection may manifest as actinomycetomas or actinomycotic granuloma, in which the fibrotic component predominates (Fig 18) (39). Actinomycetomas may form solid nodular or granulomatous mass lesions, which may be found at various sites within the central nervous system. In these cases, actinomycosis is hard to suspect until there is histopathologic confirmation from the surgical specimen.

Treatment and Prognosis

Penicillin G is the drug of choice regardless of the severity and site of actinomycosis (1,2,18). Although therapy needs to be individually tailored, mild forms of the disease, particularly in the cervicofacial region, can be adequately treated with a 2-month course of an oral antimicrobial regimen without the need for surgical resection (2). In contrast, for more severe and complicated cases, combined medical and surgical treatment may be required, with a high dosage of antibiotics for a prolonged period. Although surgical intervention can be effectively performed in a bid to remove necrotic tissue, sinus tracts, abscess cavities, and infected bone, the surgery itself is often not curative without

the aid of antimicrobial treatment (2). In any case, it is of utmost importance to treat this disease for a prolonged period of time because of the poor penetration of antibiotics into the fibrotic tissues (40). According to the findings from recent studies, however, the use of the traditional intensive long-term antibiotic regimen has been questioned (41,42). Given that most actinomycotic infections are polymicrobial in nature, it is still debatable whether antibiotic therapy needs to be directed against secondary organisms that are frequently isolated along with *Actinomyces* species (1,2,18).

In general, actinomycosis shows an excellent prognosis with a low mortality if early diagnosis is made and proper treatment is given (43). However, long-term observation and follow-up after treatments are required because relapse is common. Although disseminated disease is uncommon, it could be fatal, even if correctly diagnosed and treated (11). In central nervous system actinomycosis, Smego (35) noted several risk factors for a poor prognosis, including the onset of disease more than 2 months before presentation, a lack of antibiotic treatment or surgery, and the performance of percutaneous drainage instead of surgical resection. Compared with those who have the cervicofacial, abdominal, and pelvic forms of the disease, patients with pulmonary actinomycosis may have a much worse prognosis (2,12). This might be

partly attributable to the fact that in cases with actinomycotic bacteremia, the lungs were recognized as the initial focus of infection (11,12).

Conclusions

Actinomycosis has always been a diagnostic conundrum. In the absence of a classic clinical scenario including cutaneous sinus tracts and frequent relapses after a short course of medical therapy, this unique infection is easily confused with malignant disease or chronic granulomatous disease such as tuberculosis or fungal infection. Although actinomycosis tends to show no specific radiologic findings, as well as indolent and nonspecific symptoms, some imaging features appear to be helpful for the diagnosis of actinomycosis. These features include a highly infiltrative nature, strong enhancement after contrast material administration, and suppurative necrosis within the mass. Given that actinomycosis can be cured with antibiotic therapy alone, it is axiomatic that the timely detection of actinomycotic infection offers the greatest chance for cure without unnecessary surgery. Thus, every radiologist should be familiar with the various clinical and radiologic characteristics of actinomycosis, to make the correct diagnosis.

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Imaging of Actinomycosis in Various Organs: A Comprehensive Review

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Actinomycotic infection should be a diagnostic consideration in the following three clinical circumstances: (a) a series of manifestations including chronicity, extensive propagation across tissue planes, and firm to hard masslike features, which are frequently confused with malignant disease; (b) drainage of an abscess by a sinus tract, which may spontaneously close and reform elsewhere; and (c) temporary improvement after a short course of antibiotic treatment, followed by frequent relapses (1).

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This infiltrative nature may be explained by the action of the proteolytic enzymes of the organism (16).

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At CT, pulmonary actinomycotic infection usually appears as chronic segmental airspace consolidation containing necrotic low-attenuation areas with peripheral enhancement, often accompanied by adjacent pleural thickening (Fig 7) (19,22).

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In cases of actinomycotic disease involving the bowel, the common CT finding is concentric wall thickening, accompanied by a cystic or solid mass in the vicinity of the affected bowel segment (Fig 12) (27).

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Penicillin G is the drug of choice regardless of the severity and site of actinomycosis (1,2,18).